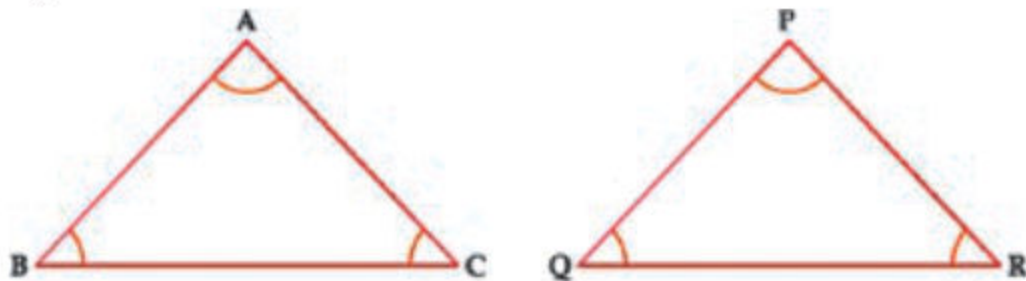


Introduction

A triangle has six elements, three sides and three angles. If we are given two triangles ABC and PQR, we can associate their vertices to establish a (1-1) correspondence between the sides and angles of these triangles in six different ways given as under:



In the correspondence $\triangle ABC \leftrightarrow \triangle PQR$ it means

- (i) $\angle A \leftrightarrow \angle P$ ($\angle A$ corresponds to $\angle P$).
- (ii) $\angle B \leftrightarrow \angle Q$ ($\angle B$ corresponds to $\angle Q$).
- (iii) $\angle C \leftrightarrow \angle R$ ($\angle C$ corresponds to $\angle R$).
- (iv) $\overline{AB} \leftrightarrow \overline{PQ}$ ($\overline{AB}$ corresponds to $\overline{PQ}$).
- (v) $\overline{BC} \leftrightarrow \overline{QR}$ ($\overline{BC}$ corresponds to $\overline{QR}$).
- (vi) $\overline{CA} \leftrightarrow \overline{RP}$ ($\overline{CA}$ corresponds to $\overline{RP}$).

9.1 Congruent triangles

"Sameness of size and shape" in the mathematics called congruence.

Consider two cars having different colours and positions as shown in the adjacent figure. But they have same size and shape. These two cars are said to be congruent. If we keep the picture of one car on the other car then they will overlap with each other.



ACTIVITY

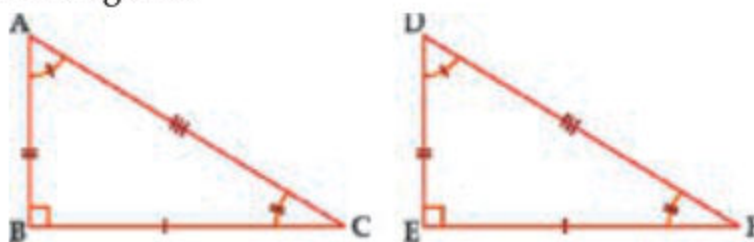
Exploration

Can you identify any congruent figures or objects in your classroom or school?

Make a list of these congruent figures by drawing or taking photos.



Two triangles are said to be congruent if their corresponding angles and sides are congruent.
Let's see the figures.



These two triangles ABC and DEF are congruent and written as:
 $\triangle ABC \cong \triangle DEF$

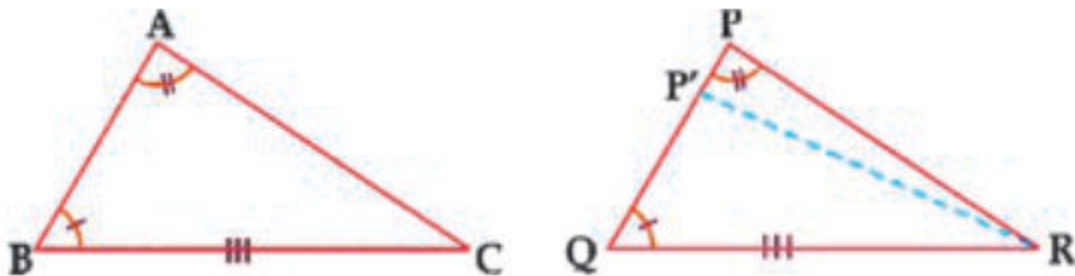
The $\triangle ABC$ and $\triangle DEF$ having their corresponding sides and angles are equal in measure.

Note: Following results are useful.

- (i) Identity congruence i.e $\triangle ABC \cong \triangle ABC$.
- (ii) Symmetric property i.e $\triangle ABC \cong \triangle PQR$ then $\triangle PQR \cong \triangle ABC$.
- (iii) Transitive property of congruence, if $\triangle ABC \cong \triangle PQR$ and $\triangle PQR \cong \triangle DEF$, then $\triangle ABC \cong \triangle DEF$.

Theorem 9.1.1 (A.S.A. $\cong$ A.S.A.)

In any correspondence of two triangles, if one side and any two angles of one triangle are congruent to the corresponding side and angles of the other, the two triangles are congruent.



Given:

In $\triangle ABC \leftrightarrow \triangle PQR$, then
 $\angle B \cong \angle Q$, $m\overline{BC} \cong m\overline{QR}$,
and $\angle A \cong \angle P$.

To prove:

$\triangle ABC \cong \triangle PQR$



Construction:

Suppose, $\overline{AB} \not\cong \overline{PQ}$ then take a point P' on $\overline{PQ}$ such that $\overline{AB} \cong \overline{P'Q}$.

Join P' to R .

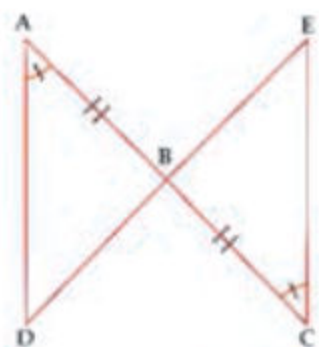
Proof:

Statements	Reasons
1. In $\triangle ABC \leftrightarrow \triangle PQR$ i. $\angle A \cong \angle P$ ii. $\angle B \cong \angle Q$	1. Correspondence of two $\triangle s$ i. Given ii. Given
2. $\therefore \angle C \cong \angle R$	2. Two corresponding angles of both triangles are congruent.
3. If $\overline{BA} \not\cong \overline{QP}$, take a point P' on $\overline{QP}$ (or $\overline{QP}$ produced) such that: $\overline{QP'} \cong \overline{BA}$	3. Assumption
4. In $\triangle ABC \leftrightarrow \triangle P'QR$ i. $\overline{BC} \cong \overline{QR}$ ii. $\angle B \cong \angle Q$ iii. $\overline{BA} \cong \overline{QP'}$	4. Correspondence of two $\triangle s$ i. Given ii. Given iii. By supposition
5. $\therefore \triangle ABC \cong \triangle P'QR$	5. S.A.S postulate
6. $\therefore \angle C \cong \angle QRP'$	6. Corresponding $\angle s$ of congruent $\triangle s$.
7. But $\angle C \cong \angle QRP$	7. Proved in 2 (above).
8. $\therefore \angle QRP' \cong \angle QRP$	8. Transitive property of congruence
9. This is possible only when points P' and P coincide and $\overline{RP'} \cong \overline{RP}$	9. By angle construction postulate
10. Hence $\overline{BA} \cong \overline{QP}$	10. As P and P' coincide.
11. In $\triangle ABC \leftrightarrow \triangle PQR$ i. $\overline{BC} \cong \overline{QR}$ ii. $\angle B \cong \angle Q$ iii. $\overline{BA} \cong \overline{QP}$	11. Correspondence of two $\triangle s$ i. Given ii. Given iii. Proved above
12. $\therefore \triangle ABC \cong \triangle PQR$	12. S.A.S Postulate.

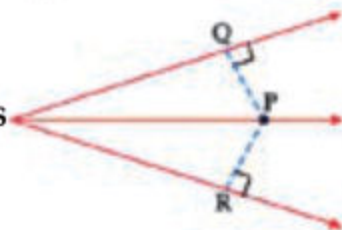
Q.E.D.

Exercise 9.1

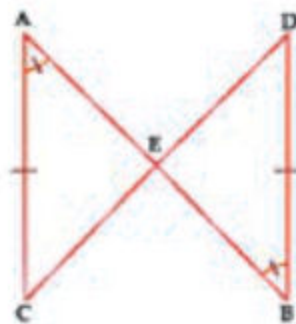
1. In the adjacent figure, $m\overline{AB} = m\overline{CB}$ and $\angle A \cong \angle C$ prove that $\triangle ABD \cong \triangle CBE$



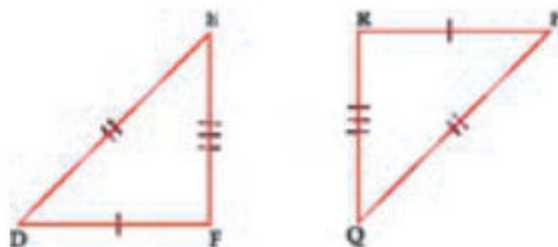
2. From a point on the line bisector of an angle, perpendiculars are drawn to the arms of the angle. Prove that these perpendiculars are equal in measure.



3. In the given figure, we have, $\triangle ACE \cong \triangle BDE$, such that $m\angle A = (3x + 1)^\circ$, $m\angle B = (x + 35)^\circ$, $m\angle AEC = (3y - 2)^\circ$ and $m\angle DEB = (y + 8)^\circ$. Find the values of x and y .



4. In the given figure, $\triangle DEF \cong \triangle PQR$, such that: $m\overline{DE} = (6x + 1)\text{cm}$, $m\overline{EF} = 8\text{cm}$, and $m\overline{RQ} = (5y - 7)\text{cm}$ and $m\overline{PQ} = (10x - 19)\text{cm}$. Find the values of x and y .



Theorem 9.1.2

If two angles of a triangle are congruent, then the sides opposite to them are also congruent.

Given:

In $\triangle ABC$,

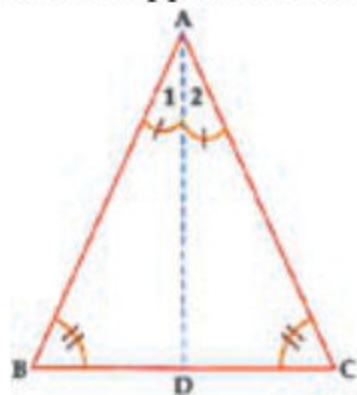
We have, $\angle B \cong \angle C$

To prove:

$$\overline{AC} \cong \overline{AB}$$

Construction: Draw $\overline{AD}$ the bisector of $\angle A$, meeting $\overline{BC}$ at point D.

Proof:



Statement	Reason
In $\triangle ADB \leftrightarrow \triangle ADC$	
i. $\angle B \cong \angle C$	i. Given
ii. $\angle 1 \cong \angle 2$	ii. Construction
iii. $\overline{AD} \cong \overline{AD}$	iii. Common side of both $\triangle s$ (Identity congruence)
$\therefore \triangle ADB \cong \triangle ADC$	A.S.A $\cong$ A.S.A
$\therefore \overline{AB} \cong \overline{AC}$	Corresponding sides of congruent $\triangle s$

Q.E.D

Exercise 9.2

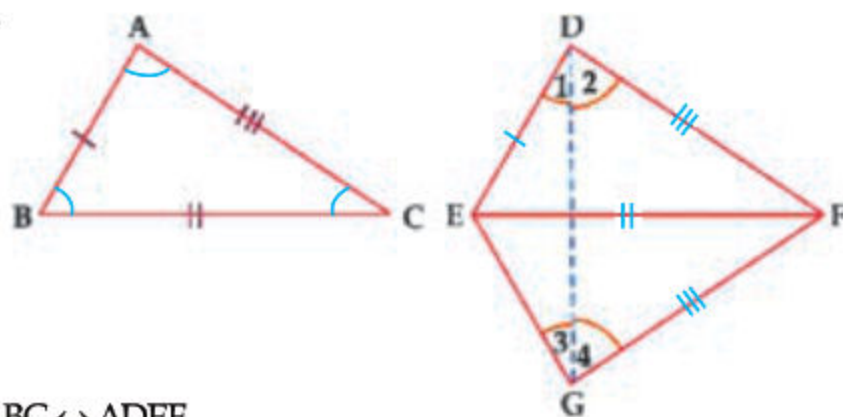
1. ABC is a triangle in which $m\angle A = 35^\circ$ and $m\angle B = 100^\circ$, $\overline{BD} \perp \overline{AC}$. Prove that $\triangle BDC$ is an isosceles triangle.
2. If the bisector of an angle of a triangle is perpendicular to its opposite side, then prove that triangle is an isosceles triangle.
3. ABC is a triangle in which $m\angle B = 45^\circ$ and $\overline{CD} \perp \overline{AB}$. Prove that $\triangle DBC$ is an isosceles $\triangle$.



Theorem 9.1.3

In a correspondence of two triangles, if three sides of one triangle are congruent to the corresponding three sides of the other, then the two triangles are congruent.

Proof:



Given:

$$\text{In } \triangle ABC \leftrightarrow \triangle DEF$$

$$\overline{AB} \cong \overline{DE}, \overline{BC} \cong \overline{EF} \text{ and } \overline{CA} \cong \overline{FD}$$

To prove that: $\triangle ABC \cong \triangle DEF$

Construction: Suppose $\overline{BC}$ is the greatest of all the three sides of $\triangle ABC$. Construct $\triangle GEF$ such that:

- i. Point G is on the opposite side of point D.
 - ii. $\angle FEG \cong \angle B$
 - iii. $\overline{EG} \cong \overline{BA}$
- Join D and G.

Proof:

Statements	Reasons
1. In $\triangle ABC \leftrightarrow \triangle GEF$	1. Correspondence of two $\triangle s$
i. $\overline{BC} \cong \overline{EF}$	i. Given
ii. $\angle B \cong \angle GEF$	ii. Construction
iii. $\overline{BA} \cong \overline{GE}$	iii. Construction
2. $\therefore \triangle ABC \cong \triangle GEF$	2. S.A.S. postulate.
3. $\therefore \overline{AC} \cong \overline{GF}$ and $\angle A \cong \angle G$	3. By the congruence of triangles.
4. But $\overline{DF} \cong \overline{AC}$	4. Given
5. $\therefore \overline{GF} \cong \overline{DF}$	5. Transitive property.
6. $\therefore$ In $\triangle DEG$, $m\angle 1 = m\angle 3$	6. Opposite sides congruent $\overline{EG} \cong \overline{BA} \cong \overline{ED}$



7. Similarly, in $\triangle GFD$, $m\angle 2 = m\angle 4$

8. $\therefore m\angle 1 + m\angle 2 = m\angle 3 + m\angle 4$

9. or $m\angle D = m\angle G$

10. But $m\angle G = m\angle A$

11. $\therefore m\angle A = m\angle D$

12. In $\triangle ABC \leftrightarrow \triangle DEF$

i. $\overline{AB} \cong \overline{DE}$

ii. $\angle A \cong \angle D$

iii. $\overline{AC} \cong \overline{DF}$

13. $\therefore \triangle ABC \cong \triangle DEF$

7. $\overline{DF} \cong \overline{GF}$

8. Addition property of equation

9. $m\angle 1 + m\angle 2 = m\angle D$

$m\angle 3 + m\angle 4 = m\angle G$

10. Proved in (3) above

11. Transitive property in (3)

12. Correspondence of two $\triangle$ s

i. Given

ii. Proved above

iii. Given

13. S.A.S Postulate

Q.E.D

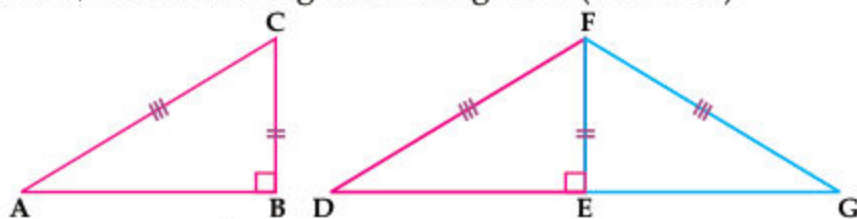
Corollary: The angles of an equilateral triangle are also equal in measurement.

Exercise 9.3

1. ABC is an isosceles triangle. D is the mid-point of base $\overline{BC}$. Prove that $\overline{AD}$ bisects $\angle A$ and $\overline{AD} \perp \overline{BC}$.
2. ABC and DBC are two isosceles triangles on the same side of a common base $\overline{BC}$. Prove that $\overline{AD}$ is the right bisector of $\overline{BC}$.
3. PQRS is a square. X, Y and Z are the mid-points of $\overline{PQ}$, $\overline{QR}$ and $\overline{RS}$ respectively. Prove that $\triangle PXY \cong \triangle SZY$.
4. Prove that, in an equilateral triangle any two median are congruent.


Theorem 9.1.4

If in the correspondence of two right-angled triangles, the hypotenuse and one side of one triangle are congruent to the hypotenuse and the corresponding side of the other, then the triangles are congruent (H.S. $\cong$ H.S.).



Given: In correspondence
 $\triangle ABC \leftrightarrow \triangle DEF$
 $\angle B \cong \angle E$ (rt $\angle$ s) $\overline{AC} \cong \overline{DF}$ (Hyp) and $\overline{BC} \cong \overline{EF}$

To Prove: $\triangle ABC \cong \triangle DEF$

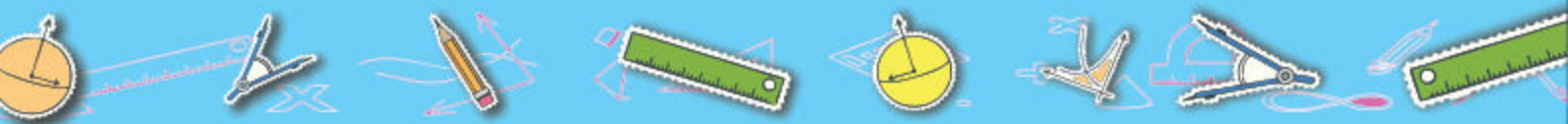
Construction: Produce $\overline{DE}$ to point G such that $\overline{EG} \cong \overline{AB}$. Then join F and G.

Proof:

Statements	Reasons
$m\angle DEF + m\angle GEF = 180^\circ$	Supplement postulate
But $m\angle DEF = 90^\circ$	Given
$\therefore m\angle GEF = 90^\circ$	$\therefore 180^\circ - 90^\circ = 90^\circ$
In $\triangle GEF \leftrightarrow \triangle ABC$	
i. $\overline{GE} \cong \overline{AB}$	Construction
ii. $\angle GEF \cong \angle ABC$	Each is right angle
iii. $\overline{EF} \cong \overline{BC}$	Given
$\therefore \triangle GEF \cong \triangle ABC$	S.A.S. $\cong$ S.A.S.
$\therefore \overline{FG} \cong \overline{AC}$ and $\angle G \cong \angle A$	By the congruence of $\triangle$ s.
$\therefore \overline{AC} \cong \overline{DF}$	$\therefore \overline{AC} \cong \overline{DF}$ (Given)
In $\triangle DFG$, $\angle D \cong \angle G$	Opposite sides congruents
$\therefore \angle D \cong \angle A$	Each is congruent to $\angle G$
In $\triangle ABC \leftrightarrow \triangle DEF$	
i. $\angle A \cong \angle D$	i. Proved
ii. $\angle ABC \cong \angle DEF$	ii. rt $\triangle$ s
iii. $\overline{AC} \cong \overline{DF}$	iii. Given
$\therefore \triangle ABC \cong \triangle DEF$	A.A.S. $\cong$ A.A.S.

Q.E.D

Note: Theorem 9.1.4 can be proved by S.A.S postulate.



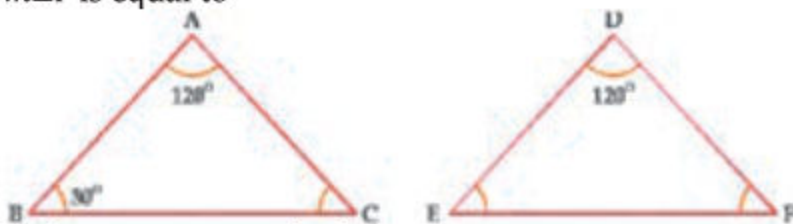
Exercise 9.4

1. Prove that:
The perpendiculars from the vertices of the base to opposite sides of an isosceles triangle are congruent. (Hint: Medians and altitudes of the triangle are congruent)
2. Prove that, if the bisector of an angle of a triangle bisects its opposite side, then the triangle will be an isosceles triangle.
3. Prove that the median bisecting the base of an isosceles triangle bisects the vertical angle and is perpendicular to the base.
4. Prove that if three altitudes of a triangle are congruent, then the triangle is equilateral.

Review Exercise 9

1. If $\triangle ABC \cong \triangle DEF$, $m\angle F$ is equal to

- A. 90°
- B. 60°
- C. 30°
- D. 20°



2. Identify true and false statement in the following:

- (i) The sum of the measure of all angles in a quadrilateral is 360° .
- (ii) The sum of the measure of all angles in a triangle is 270° .
- (iii) In an equilateral triangle, angles are of the same measurement.
- (iv) There are two right angles in a triangle.
- (v) In an isosceles triangles, corresponding angles and corresponding sides are equal in measure.

3. Fill in the blanks to make the sentences true sentences:

- (i) In $\triangle ABC \leftrightarrow \triangle DEF$, then $\overline{AC}$ corresponds to _____.
- (ii) In $\triangle KLM \leftrightarrow \triangle PQR$, then $\angle MKL$ corresponds to _____.
- (iii) In an isosceles triangle, the base angle are _____.
- (iv) If the measure of each of the angles of a triangle is 60° , then the triangle is _____.
- (v) In a right-angled triangle, side opposite to right angle is called _____.
- (vi) The sum of the measures of acute angle of a right triangle is _____.



4. Encircle the corresponding letters a,b,c or d for correct answer:
- Which of the following is not a sufficient condition for congruence of two triangles?

(a) $A.S.A \cong A.S.A$	(b) $H.S \cong H.S$
(c) $S.A.A \cong S.A.A$	(d) $A.A.A \cong A.A.A$
 - In $\triangle ABC$, if $\angle A \cong \angle B$, then the bisector of ____ angle divides the triangle into congruent triangles:

(a) $\angle A$	(b) $\angle B$
(c) $\angle C$	(d) any one of its angles.
 - The diagonal of ____ does not divide it into two congruent triangles:

(a) Rectangle	(b) Trapezium
(c) Parallelogram	(d) Square
 - How many acute angles are there in an acute angled triangle?

(a) 1	(b) 2
(c) 3	(d) not more than 2.



Summary

In this unit we stated and proved the following theorem:

- ◆ In any correspondence of two triangles, if one side and any two angles of one triangle are congruent to the corresponding side and angles of the other, the two triangles are congruent. ($A.S.A \cong A.S.A$)
- ◆ If two angles of a triangle are congruent, then the side opposite to them are also congruent.
- ◆ In the correspondence of two triangles, if three sides of two triangles are congruent to the corresponding three sides of other, then the two triangles are congruent ($S.S.S \cong S.S.S$).
- ◆ If in the correspondence of the two right-angled triangles the hypotenuse and one side of one triangle are congruent to the hypotenuse and the corresponding side of other, then the triangles are congruent. ($H.S \cong H.S$).
- ◆ Two triangles are said to be congruent, if there exists a correspondence between them such that all the corresponding sides and triangles are congruent. ($S.S.S$).



Unit

10

• Weightage = 7%

PARALLELOGRAM AND TRIANGLES

Student Learning Outcomes (SLOs)

After completing this unit, students will be able to:

- ◆ Understand the following theorems along with their corollaries and apply them to solve allied problems.
 - a) In a parallelogram:
 - ◆ The opposite sides are congruent,
 - ◆ The opposites angles are congruent,
 - ◆ The diagonals bisect each other.
 - b) If two opposite sides of a quadrilateral are congruent and parallel, it is a parallelogram.
 - c) The line segments joining the midpoints of two sides of a triangle, is parallel to the third side and it is equal to one half of its length.
 - d) The medians of a triangle are concurrent and their point of concurrency is the point of trisection of each median.
 - e) If three or more parallel lines make congruent intercepts on the transversal, they also intercept congruent segments on any other line that cuts them.